Reproduction

Mark scheme and worked solutions for 60 questions: 30 multiple choice, 30 structured.

1 ? Part 1

Core | Asexual reproduction | [1]

Answer: B

Explanation

Asexual reproduction uses one parent and produces genetically identical offspring. Neither gamete fusion nor the mixing of two parents' genes occurs.

2 ? Part 1

Core | Asexual reproduction | [1]

Answer: D

Explanation

Division from one parent without gamete fusion is asexual reproduction. Pollination transfers pollen; fertilisation and sexual reproduction involve gamete nuclei.

3 ? Part 1

Supplement | Asexual reproduction | [1]

Answer: A

Explanation

Cloning retains the parent's genetic combination. It does not create broad variation or guarantee disease resistance, and it does not require pollination.

4 ? Part 1

Core | Sexual reproduction | [1]

Answer: C

Explanation

Sexual reproduction involves fusion of gamete nuclei. Growth, cell multiplication and production of offspring also occur in asexual reproduction.

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5 ? Part 1

Core | Sexual reproduction | [1]

Answer: D

Explanation

Fertilisation is nuclear fusion. Landing on the stigma is pollination, while tube growth enables the nuclei to meet but is not itself fertilisation.

6 ? Part 1

Supplement | Sexual reproduction | [1]

Answer: B

Explanation

Two haploid sets of 12 combine, giving 12 + 12 = 24 chromosomes. Fertilisation adds sets rather than halving or doubling each set separately.

7 ? Part 1

Core | Sexual reproduction in plants | [1]

Answer: C

Explanation

The anther produces pollen. The stigma receives it, the ovary contains ovules, and petals may attract pollinators.

8 ? Part 1

Core | Sexual reproduction in plants | [1]

Answer: A

Explanation

A spiky surface helps pollen attach to an insect. Light, smooth grains are more characteristic of wind dispersal; preventing contact would hinder insect pollination.

9 ? Part 1

Core | Sexual reproduction in plants | [1]

Answer: B

Explanation

Pollination transfers pollen from an anther to a stigma. Movement to an ovule occurs later, involving a pollen tube rather than transfer of whole grains.

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10 ? Part 1 Core Sexual reproduction in plants [1]

Answer: C

Explanation

Exposed anthers release pollen into moving air and a feathery stigma traps airborne grains. Nectar, scent and showy petals chiefly attract animals.

11 ? Part 1 Core Sexual reproduction in plants [1]

Answer: D

Explanation

Water is required for germination. Light is not a universal requirement, so darkness alone does not prevent all viable seeds from germinating.

12 ? Part 1 Supplement Sexual reproduction in plants [1]

Answer: A

Explanation

Both flowers belong to the same plant, so this is self-pollination. Cross-pollination requires a different plant of the same species.

13 ? Part 1 Supplement Sexual reproduction in plants [1]

Answer: C

Explanation

Cross-pollination transfers pollen to a flower on another plant of the same species. Neither within-plant transfer nor seed movement fits this definition.

14 ? Part 1 Supplement Sexual reproduction in plants [1]

Answer: A

Explanation

Cross-pollination requires transfer between plants, so reproduction can become vulnerable to failure of that transfer. It does not abolish fertilisation or embryo formation.

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15 ? Part 1 Supplement Sexual reproduction in plants [1]

Answer: B

Explanation

The tube grows from the stigma through the style towards an ovule, carrying the male nucleus. The other routes do not deliver it to the female nucleus.

16 ? Part 1 Supplement Asexual reproduction [4]

Mark scheme

(a) **[2]**

- The plants have the same genetic information for relevant inherited features. (1)
- They are therefore likely to produce similar crop characteristics under the same conditions. (1)

(b) **[2]**

- The plants have little genetic variation. (1)
- If the cloned genotype is susceptible, many plants may be affected together. (1)

Worked solution

(a) The plants have the same genetic information for relevant inherited features. They are therefore likely to produce similar crop characteristics under the same conditions.

(b) The plants have little genetic variation. If the cloned genotype is susceptible, many plants may be affected together.

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17 ? Part 1 Supplement Asexual reproduction [4]

Mark scheme

(a) **[2]**

- Only one parent is required, so no mate is needed. (1)
- Offspring can be produced without the delay of finding another reproductive individual. (1)

(b) **[2]**

- The offspring are genetically identical apart from mutation. (1)
- There may be no individuals with inherited features suited to the changed conditions. (1)

Worked solution

(a) Only one parent is required, so no mate is needed. Offspring can be produced without the delay of finding another reproductive individual.

(b) The offspring are genetically identical apart from mutation. There may be no individuals with inherited features suited to the changed conditions.

18 ? Part 1 Core Sexual reproduction [4]

Mark scheme

(a) **[2]**

- Fertilisation is fusion of the nuclei of two gametes. (1)
- The resulting cell is a zygote. (1)

(b) **[2]**

- Each offspring receives genetic information from both gametes. (1)
- Offspring are genetically different from one another. (1)

Worked solution

(a) Fertilisation is fusion of the nuclei of two gametes. The resulting cell is a zygote.

(b) Each offspring receives genetic information from both gametes. Offspring are genetically different from one another.

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19 ? Part 1

Supplement | Sexual reproduction | [4]

Mark scheme

(a) **[2]**

- A gamete has 9 chromosomes. (1)
- A gamete is haploid and has one set, half the diploid body-cell number. (1)

(b) **[2]**

- The zygote has 18 chromosomes. (1)
- The two gamete nuclei contribute 9 chromosomes each during fusion. (1)

Worked solution

(a) A gamete has 9 chromosomes. A gamete is haploid and has one set, half the diploid body-cell number.

(b) The zygote has 18 chromosomes. The two gamete nuclei contribute 9 chromosomes each during fusion.

20 ? Part 1

Supplement | Sexual reproduction | [4]

Mark scheme

(a) **[2]**

- Sexual reproduction produces genetic variation among offspring. (1)
- Some offspring may possess a combination of alleles better suited to the new climate. (1)

(b) **[2]**

- The offspring may not retain all desirable parental characteristics. (1)
- Time and resources are needed for sexual reproduction and subsequent selection. (1)

Worked solution

(a) Sexual reproduction produces genetic variation among offspring. Some offspring may possess a combination of alleles better suited to the new climate.

(b) The offspring may not retain all desirable parental characteristics. Time and resources are needed for sexual reproduction and subsequent selection.

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21 ? Part 1 Supplement Sexual reproduction [4]

Mark scheme

(a) **[2]**

- Some individuals may have inherited features that help them avoid the predator. (1)
- These individuals are more likely to survive and reproduce. (1)

(b) **[2]**

- Individuals must obtain compatible gametes, normally by finding a mate. (1)
- Separation reduces the chance of mating and therefore successful fertilisation. (1)

Worked solution

(a) Some individuals may have inherited features that help them avoid the predator. These individuals are more likely to survive and reproduce.

(b) Individuals must obtain compatible gametes, normally by finding a mate. Separation reduces the chance of mating and therefore successful fertilisation.

22 ? Part 1 Core Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- The stigma receives pollen grains. (1)
- The ovary contains ovules. (1)

(b) **[2]**

- The damaged stigma may receive less pollen, reducing successful fertilisation. (1)
- The intact anthers can still produce pollen grains. (1)

Worked solution

(a) The stigma receives pollen grains. The ovary contains ovules.

(b) The damaged stigma may receive less pollen, reducing successful fertilisation. The intact anthers can still produce pollen grains.

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23 ? Part 1 Core Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- A is insect-pollinated because its nectar or colour attracts insects. (1)
- B is wind-pollinated because exposed hanging anthers release pollen into air. (1)

(b) **[2]**

- A is likely to have stickier or spikier pollen than B. (1)
- B is likely to produce smaller, lighter pollen grains than A. (1)

Worked solution

(a) A is insect-pollinated because its nectar or colour attracts insects. B is wind-pollinated because exposed hanging anthers release pollen into air.

(b) A is likely to have stickier or spikier pollen than B. B is likely to produce smaller, lighter pollen grains than A.

24 ? Part 1 Core Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- Pollination is transfer of pollen grains from anther to stigma. (1)
- Fertilisation is fusion of a pollen nucleus with a nucleus in an ovule. (1)

(b) **[2]**

- The pollen may be incompatible with the receiving flower. (1)
- The pollen grain may fail to germinate or form a functioning pollen tube. (1)

Worked solution

(a) Pollination is transfer of pollen grains from anther to stigma. Fertilisation is fusion of a pollen nucleus with a nucleus in an ovule.

(b) The pollen may be incompatible with the receiving flower. The pollen grain may fail to germinate or form a functioning pollen tube.

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25 ? Part 1 Core Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- The condition being tested is water availability. (1)
- More seeds are expected to germinate in A than in B. (1)

(b) **[2]**

- The temperature must be the same. (1)
- The availability of oxygen must be the same. (1)

Worked solution

(a) The condition being tested is water availability. More seeds are expected to germinate in A than in B.

(b) The temperature must be the same. The availability of oxygen must be the same.

26 ? Part 1 Core Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- Seeds in air germinate while those without oxygen do not. (1)
- Oxygen is required for germination under these conditions. (1)

(b) **[2]**

- Use seeds of the same species and similar viability. (1)
- Provide the same water availability in both containers. (1)

Worked solution

(a) Seeds in air germinate while those without oxygen do not. Oxygen is required for germination under these conditions.

(b) Use seeds of the same species and similar viability. Provide the same water availability in both containers.

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27 ? Part 1 Supplement Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- X to Y is self-pollination because both flowers are on one plant. (1)
- X to Z is cross-pollination because the flowers are on different plants of the same species. (1)

(b) **[2]**

- The pollen starts on an anther. (1)
- It is received by a stigma. (1)

Worked solution

(a) X to Y is self-pollination because both flowers are on one plant. X to Z is cross-pollination because the flowers are on different plants of the same species.

(b) The pollen starts on an anther. It is received by a stigma.

28 ? Part 1 Supplement Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- Q is more likely to have a wider range of combinations. (1)
- Cross-pollination combines genetic information from different plants. (1)

(b) **[2]**

- P may still reproduce if self-pollination can occur without insects. (1)
- Q may produce fewer seeds because pollen transfer between plants is reduced. (1)

Worked solution

(a) Q is more likely to have a wider range of combinations. Cross-pollination combines genetic information from different plants.

(b) P may still reproduce if self-pollination can occur without insects. Q may produce fewer seeds because pollen transfer between plants is reduced.

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29 ? Part 1 Supplement Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- A pollen tube grows through the style towards an ovule. (1)
- The male nucleus travels along the pollen tube into the ovule. (1)

(b) **[2]**

- The male nucleus fuses with a nucleus in the ovule. (1)
- This forms a zygote. (1)

Worked solution

(a) A pollen tube grows through the style towards an ovule. The male nucleus travels along the pollen tube into the ovule.

(b) The male nucleus fuses with a nucleus in the ovule. This forms a zygote.

30 ? Part 1 Core Sexual reproduction in plants [4]

Mark scheme

(a) **[2]**

- Exposed anthers allow pollen to be released into the wind. (1)
- Large feathery stigmas increase the chance of trapping airborne pollen. (1)

(b) **[2]**

- Oxygen must be available. (1)
- The temperature must be suitable for the species. (1)

Worked solution

(a) Exposed anthers allow pollen to be released into the wind. Large feathery stigmas increase the chance of trapping airborne pollen.

(b) Oxygen must be available. The temperature must be suitable for the species.

1 ? Part 2 Core Sexual reproduction in humans [1]

Answer: B

Explanation

The acrosome contains enzymes that help penetration. The flagellum provides movement and the nucleus carries genetic information.

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2 ? Part 2

Core | Sexual reproduction in humans | [1]

Answer: D

Explanation

The jelly coat changes after fertilisation, preventing further sperm entry. Nutrient loss and new motility structures are not the protective response.

3 ? Part 2

Core | Sexual reproduction in humans | [1]

Answer: A

Explanation

Eggs are larger and have energy stores; sperm are small, motile and produced in much greater numbers.

4 ? Part 2

Core | Sexual reproduction in humans | [1]

Answer: C

Explanation

The zygote divides to form an embryo, which implants in the uterine lining. Anthers are plant structures and implantation normally occurs in the uterus.

5 ? Part 2

Supplement | Sexual reproduction in humans | [1]

Answer: D

Explanation

Carbon dioxide and urea are fetal waste products. Oxygen and dissolved nutrients normally move towards the fetus.

6 ? Part 2

Supplement | Sexual reproduction in humans | [1]

Answer: B

Explanation

The placenta allows exchange and is not a complete barrier to toxins and pathogens. Maternal and fetal blood are separate circulations.

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7 ? Part 2

Core Sex hormones in humans [1]

Answer: C

Explanation

Testosterone promotes male secondary sexual characteristics. Oestrogen promotes female secondary sexual characteristics; the other listed hormones have different reproductive roles.

8 ? Part 2

Core Sex hormones in humans [1]

Answer: A

Explanation

Menstruation involves shedding the uterine lining. Release of an egg is ovulation, a separate event in an ovary.

9 ? Part 2

Supplement Sex hormones in humans [1]

Answer: B

Explanation

The rise in LH triggers ovulation. The other hormones do not provide this menstrual-cycle signal.

10 ? Part 2

Supplement Sex hormones in humans [1]

Answer: C

Explanation

The corpus luteum in the ovary produces progesterone after ovulation. The pituitary releases FSH and LH rather than progesterone.

11 ? Part 2

Core Sexually transmitted infections [1]

Answer: D

Explanation

STIs are transmitted through sexual contact. Inherited disorders and nutritional conditions are not infections, and coughing alone does not define an STI.

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12 ? Part 2 Core Sexually transmitted infections [1]

Answer: A

Explanation

HIV is a virus and infection may progress to AIDS. AIDS is a condition, not a pathogen, and many STIs have other causes.

13 ? Part 2 Core Sexually transmitted infections [1]

Answer: C

Explanation

Contaminated needles can transfer infected blood. Ordinary social contact such as handshaking or sharing desks does not transmit HIV.

14 ? Part 2 Core Sexually transmitted infections [1]

Answer: A

Explanation

A condom provides a barrier to infected body fluids. Antibiotics do not prevent HIV, a viral infection; the other actions do not provide this barrier.

15 ? Part 2 Core Sexually transmitted infections [1]

Answer: B

Explanation

HIV can be transmitted in breast milk. It is an infectious virus rather than an allele, and airborne contact and clean objects are not usual routes.

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16 ? Part 2

Core | Sexual reproduction in humans | [4]

Mark scheme

(a) **[2]**

- The sperm nucleus fuses with the egg nucleus. (1)
- The cell formed is a zygote. (1)

(b) **[2]**

- The flagellum propels the sperm. (1)
- Mitochondria release energy by respiration for movement. (1)

Worked solution

(a) The sperm nucleus fuses with the egg nucleus. The cell formed is a zygote.

(b) The flagellum propels the sperm. Mitochondria release energy by respiration for movement.

17 ? Part 2

Core | Sexual reproduction in humans | [4]

Mark scheme

(a) **[2]**

- It supplies nutrients for early development. (1)
- The stored nutrients can be used in respiration to release energy. (1)

(b) **[2]**

- Sperm are motile whereas eggs do not swim using a flagellum. (1)
- Sperm are produced in much greater numbers than eggs. (1)

Worked solution

(a) It supplies nutrients for early development. The stored nutrients can be used in respiration to release energy.

(b) Sperm are motile whereas eggs do not swim using a flagellum. Sperm are produced in much greater numbers than eggs.

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18 ? Part 2

Core | Sexual reproduction in humans | [4]

Mark scheme

(a) **[2]**

- The zygote divides repeatedly. (1)
- The embryo becomes a ball of cells. (1)

(b) **[2]**

- Implantation occurs in the lining of the uterus. (1)
- The embryo becomes embedded in that lining. (1)

Worked solution

(a) The zygote divides repeatedly. The embryo becomes a ball of cells.

(b) Implantation occurs in the lining of the uterus. The embryo becomes embedded in that lining.

19 ? Part 2

Supplement | Sexual reproduction in humans | [4]

Mark scheme

(a) **[2]**

- Glucose is transferred. (1)
- Amino acids are transferred. (1)

(b) **[2]**

- Carbon dioxide moves towards the mother. (1)
- Urea moves towards the mother. (1)

Worked solution

(a) Glucose is transferred. Amino acids are transferred.

(b) Carbon dioxide moves towards the mother. Urea moves towards the mother.

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20 ? Part 2

Supplement | Sexual reproduction in humans | [4]

Mark scheme

(a) **[2]**

- The placenta is a site of exchange between maternal and fetal blood. (1)
- Blood in the umbilical cord carries exchanged substances between placenta and fetus. (1)

(b) **[2]**

- Some toxins can cross the placenta into fetal blood. (1)
- The toxin can therefore reach fetal tissues and affect development. (1)

Worked solution

(a) The placenta is a site of exchange between maternal and fetal blood. Blood in the umbilical cord carries exchanged substances between placenta and fetus.

(b) Some toxins can cross the placenta into fetal blood. The toxin can therefore reach fetal tissues and affect development.

21 ? Part 2

Core | Sexual reproduction in humans | [4]

Mark scheme

(a) **[2]**

- The acrosome normally provides enzymes. (1)
- Reduced enzyme release makes penetration of the egg coverings less likely. (1)

(b) **[2]**

- Energy stores support early development. (1)
- The jelly coat changes after fertilisation to prevent additional sperm entering. (1)

Worked solution

(a) The acrosome normally provides enzymes. Reduced enzyme release makes penetration of the egg coverings less likely.

(b) Energy stores support early development. The jelly coat changes after fertilisation to prevent additional sperm entering.

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22 ? Part 2

Core | Sexual reproduction in humans | [4]

Mark scheme

(a) **[2]**

- Sperm have a flagellum whereas eggs do not. (1)
- Eggs have much larger nutrient stores and cytoplasm than sperm. (1)

(b) **[2]**

- Many sperm are produced compared with relatively few eggs. (1)
- Sperm swim using their flagella whereas eggs do not swim independently. (1)

Worked solution

(a) Sperm have a flagellum whereas eggs do not. Eggs have much larger nutrient stores and cytoplasm than sperm.

(b) Many sperm are produced compared with relatively few eggs. Sperm swim using their flagella whereas eggs do not swim independently.

23 ? Part 2

Core | Sex hormones in humans | [4]

Mark scheme

(a) **[2]**

- Testosterone is the hormone. (1)
- A valid example is deepening of the voice or growth of facial hair. (1)

(b) **[2]**

- Oestrogen is the hormone. (1)
- A valid example is development of breasts or widening of the hips. (1)

Worked solution

(a) Testosterone is the hormone. A valid example is deepening of the voice or growth of facial hair.

(b) Oestrogen is the hormone. A valid example is development of breasts or widening of the hips.

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24 ? Part 2

Core | Sex hormones in humans | [4]

Mark scheme

(a) **[2]**

- An egg matures in a follicle. (1)
- Ovulation releases the egg around the middle of the cycle. (1)

(b) **[2]**

- The lining thickens during the cycle. (1)
- It breaks down and is shed if pregnancy does not occur. (1)

Worked solution

(a) An egg matures in a follicle. Ovulation releases the egg around the middle of the cycle.

(b) The lining thickens during the cycle. It breaks down and is shed if pregnancy does not occur.

25 ? Part 2

Supplement | Sex hormones in humans | [4]

Mark scheme

(a) **[2]**

- The corpus luteum produces progesterone. (1)
- Progesterone maintains the uterine lining. (1)

(b) **[2]**

- It inhibits secretion of FSH and LH. (1)
- This reduces further follicle maturation and an additional ovulation. (1)

Worked solution

(a) The corpus luteum produces progesterone. Progesterone maintains the uterine lining.

(b) It inhibits secretion of FSH and LH. This reduces further follicle maturation and an additional ovulation.

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26 ? Part 2 Supplement Sex hormones in humans [4]

Mark scheme

(a) **[2]**

- FSH stimulates maturation of an ovarian follicle. (1)
- It stimulates oestrogen secretion by the developing follicle. (1)

(b) **[2]**

- Oestrogen stimulates rebuilding or thickening of the uterine lining. (1)
- A sufficiently high oestrogen concentration stimulates the LH surge that triggers ovulation. (1)

Worked solution

(a) FSH stimulates maturation of an ovarian follicle. It stimulates oestrogen secretion by the developing follicle.

(b) Oestrogen stimulates rebuilding or thickening of the uterine lining. A sufficiently high oestrogen concentration stimulates the LH surge that triggers ovulation.

27 ? Part 2 Core Sexually transmitted infections [4]

Mark scheme

(a) **[2]**

- An STI is an infection transmitted through sexual contact. (1)
- HIV is one example of an STI pathogen. (1)

(b) **[2]**

- Education can encourage people to avoid risky sexual contact and seek testing. (1)
- Condoms form a barrier that reduces transfer of infected body fluids. (1)

Worked solution

(a) An STI is an infection transmitted through sexual contact. HIV is one example of an STI pathogen.

(b) Education can encourage people to avoid risky sexual contact and seek testing. Condoms form a barrier that reduces transfer of infected body fluids.

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28 ? Part 2

Core | Sexually transmitted infections | [4]

Mark scheme

(a) **[2]**

- Screening reduces the chance of transfusing blood containing HIV. (1)
- Single-use needles prevent transfer of infected blood between patients. (1)

(b) **[2]**

- HIV may pass during birth. (1)
- HIV may pass through breast milk. (1)

Worked solution

(a) Screening reduces the chance of transfusing blood containing HIV. Single-use needles prevent transfer of infected blood between patients.

(b) HIV may pass during birth. HIV may pass through breast milk.

29 ? Part 2

Core | Sexually transmitted infections | [4]

Mark scheme

(a) **[2]**

- HIV is a virus and a pathogen. (1)
- HIV infection may lead to the condition called AIDS. (1)

(b) **[2]**

- Transmission requires a suitable route involving infected body fluids. (1)
- Ordinary contact such as shaking hands does not provide that route. (1)

Worked solution

(a) HIV is a virus and a pathogen. HIV infection may lead to the condition called AIDS.

(b) Transmission requires a suitable route involving infected body fluids. Ordinary contact such as shaking hands does not provide that route.

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30 ? Part 2

Core | Sexually transmitted infections | [4]

Mark scheme

(a) **[2]**

- Shared equipment may transfer infected blood. (1)
- Unprotected sexual contact may transfer infected sexual fluids. (1)

(b) **[2]**

- Use sterile single-use injection equipment and do not share it. (1)
- Use a condom correctly during sexual contact. (1)

Worked solution

(a) Shared equipment may transfer infected blood. Unprotected sexual contact may transfer infected sexual fluids.

(b) Use sterile single-use injection equipment and do not share it. Use a condom correctly during sexual contact.

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